

Project Plan Guidelines

Through the project proposal, both you and the instructor have agreed on the size and scope of the project. The next step is to define the project deliverables, and the specific work required to complete with timeline. The Project Plan document should include all the necessary tasks and milestones required to successfully complete the project.

Essentially, the student will break down the major project deliverables into smaller, more manageable components. For each of these, the student should outline the required activities and estimate the resources and time needed to complete them.

Below is a general list of elements the instructor will be looking for in the Project Plan: Here is a general list of things the instructor will be looking for in the proposal:

1. Background & Literature Review

Provide a concise description of the business or academic need motivating the project, explaining why the topic is important and relevant. This section should include a review of current research or industry practices related to the project, with appropriate citations from credible sources.

2. Project Aims and Objectives

Clearly state what the project intends to accomplish. Identify specific, measurable objectives and explain how this work extends, improves upon, or differs from existing tools, methods, or solutions (if any are available).

3. Expected Outcomes Outline

Describe the questions, problems, or challenges the project aims to address. Clearly explain the expected outcomes or deliverables and how they align with the stated objectives.

4. Evaluation Criteria

Define the key features and characteristics of a successful project. Explain which aspects are essential to the project's success and describe how you will evaluate whether these criteria have been met.

5. Timeline and Milestones

Outline anticipated milestones and deliverables. Provide a detailed timeline (e.g., week-by-week) that shows each major stage of the project, along with estimated completion dates. The timeline should align with the course schedule outlined in the syllabus.

Project Plan

Project name: Data Science Job Market Analysis for Curriculum Enhancement

Background:

The demand for data science professionals has grown rapidly over the past decade, driven by the increasing reliance on data-driven decision-making across industries such as finance, healthcare, technology, and manufacturing. According to the U.S. Bureau of Labor Statistics (BLS) [1], employment of data scientists is projected to grow by 36 percent from 2023 to 2033, far exceeding the average growth rate for all occupations. Similar trends have been reported by industry analyses from LinkedIn, Glassdoor, and IBM, which consistently identify data science and analytics roles as both high-growth and high-paying career paths [2][3].

In addition to strong demand, the nature of data science roles has evolved significantly. Prior research and industry reports indicate that employers increasingly seek proficiency in cloud-based platforms (e.g., AWS, Azure, Google Cloud), modern machine learning frameworks, and big data technologies, alongside traditional statistical and programming skills [4][5]. Studies have also emphasized the growing importance of soft skills—such as communication, teamwork, and business understanding—in enabling data scientists to translate technical results into actionable insights [6][7].

These shifts pose challenges for academic data science programs, which must regularly evaluate and update curricula to remain aligned with industry needs. Several studies have highlighted gaps between university curricula and employer expectations, particularly in applied skills, tool familiarity, and real-world problem-solving experience [8][9]. As a result, there is increasing interest in using empirical labor market data, such as job postings, to inform curriculum design and assessment [10][11].

This project seeks to contribute to this effort by analyzing large-scale, real-world data science job postings to identify current and emerging skill demands. By systematically extracting and analyzing skill requirements, tools, and qualifications from job market data, the project aims to generate data-driven insights to support curriculum review and enhancement for the Bachelor of Science in Data Science program at Wentworth Institute of Technology. These insights can help ensure that program learning outcomes and course content remain relevant in a rapidly evolving and competitive job market.

Project Aims and Objectives:

The aim of this project is to conduct a systematic, data-driven analysis of current data science job market trends and to develop evidence-based recommendations for enhancing the undergraduate data science curriculum. The project emphasizes rigorous data collection, analytical methodology, and critical evaluation of curriculum alignment with industry demands.

Objectives

- Acquire and preprocess large-scale job posting data from multiple employment platforms or databases using reproducible data collection methods.

- Apply advanced analytical techniques (e.g., text mining, natural language processing, clustering, or topic modeling) to identify dominant and emerging skills, tools, and qualification requirements.
- Evaluate alignment between job market demands and the existing undergraduate data science curriculum, including required courses and program learning outcomes.
- Critically assess gaps, redundancies, and emerging areas not adequately covered in the current curriculum.
- Develop data-supported curriculum enhancement recommendations, grounded in both empirical findings and relevant educational literature.
- Communicate effectively through a comprehensive graduate-level final report and a formal poster or oral presentation.

Expected Outcomes Outline:

This project is expected to produce the following outcomes:

1. Curated and Documented Job Market Dataset

A cleaned, well-documented dataset of data science job postings, including extracted skill requirements, tools, qualifications, and role characteristics, suitable for reproducible analysis.

2. Quantitative and Qualitative Analysis of Skill Demand

Empirical findings identifying high-frequency, emerging, and declining skills, technologies, and competencies in the current data science job market, supported by statistical summaries and advanced analytical methods.

3. Skill Taxonomy and Trend Mapping

A structured categorization of technical skills, tools, and professional competencies, along with trend analyses that illustrate how industry demands align with or diverge from traditional academic content.

4. Curriculum Alignment Assessment

A systematic comparison between identified job market requirements and the existing undergraduate data science curriculum, highlighting areas of strong alignment, gaps, and potential redundancies.

5. Evidence-Based Curriculum Enhancement Recommendations

A set of actionable and prioritized recommendations for curriculum enhancement, supported by data analysis results and relevant educational literature.

6. Professional-Quality Deliverables

A comprehensive graduate-level final report documenting data sources, methodology, analysis, results, and recommendations, along with a clear and concise poster or presentation suitable for academic or institutional audiences.

Evaluation Criteria:

The success of this project will be evaluated based on the following criteria:

1. Data Quality and Reproducibility

The job posting dataset is comprehensive, properly cleaned, and well documented. Data collection and preprocessing steps are clearly described and reproducible.

2. Methodological Rigor

Appropriate analytical techniques are selected and correctly applied. Methods are well justified, and assumptions and limitations are clearly discussed.

3. Depth and Validity of Analysis

The analysis produces meaningful insights into skill demand and job market trends. Results are supported by evidence, clearly interpreted, and go beyond descriptive statistics.

4. Curriculum Alignment Analysis

The comparison between job market requirements and the existing undergraduate curriculum is systematic, accurate, and well reasoned, with clear identification of gaps, overlaps, and emerging needs.

5. Quality of Curriculum Recommendations

Recommendations are actionable, data-driven, and feasible within an academic program context. Each recommendation is clearly linked to analytical findings and relevant literature.

6. Communication and Presentation

The final report is well organized, clearly written, and meets graduate-level standards. Visualizations, tables, and figures are effective and properly labeled. The final presentation or poster clearly communicates the project's objectives, methods, findings, and implications.

7. Professionalism and Project Management

Project milestones are met according to the proposed timeline. The student demonstrates consistent progress, responsiveness to feedback, and professional conduct throughout the project.

Project Plan and Timing:

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[9] R. Saltz and J. Stanton, “An introduction to data science education,” *International Journal of Data Science Education*, vol. 1, no. 1, pp. 1–16, 2017.

[10] E. Demchenko, P. Grosso, C. de Laat, and P. Membrey, “Addressing big data challenges for scientific data infrastructure,” in *Proc. IEEE Int. Conf. Cloud Computing Technology and Science*, Bristol, U.K., 2013, pp. 614–617.

[11] J. Wang, Y. Li, and L. Zhao, “Analyzing online job postings to inform curriculum design in data science,” *IEEE Access*, vol. 8, pp. 152367–152378, 2020.